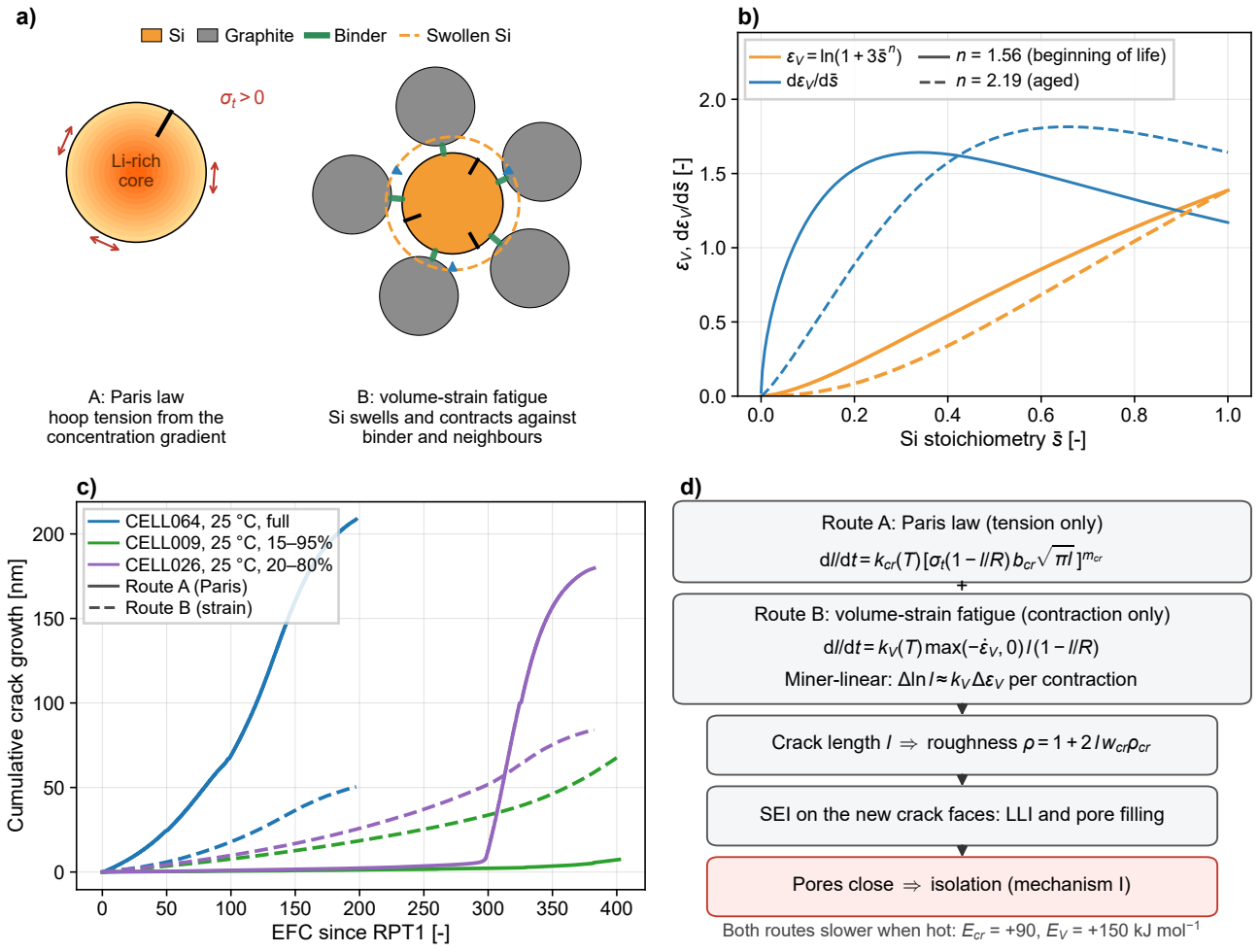

a)
Si
Graphite
Binder
Swollen Si
$\sigma_t > 0$
Li-rich core
A: Paris law
hoop tension from the
concentration gradient
B: volume-strain fatigue
Si swells and contracts against
binder and neighbours
b)
$\varepsilon_V = \ln(1 + 3\bar{s}^n)$
$d\varepsilon_V/d\bar{s}$
n = 1.56 (beginning of life)
n = 2.19 (aged)
ε_V, $d\varepsilon_V/d\bar{s}$ [-]
Si stoichiometry $\bar{s}$ [-]
c)
CELL064, 25 °C, full
CELL009, 25 °C, 15–95%
CELL026, 25 °C, 20–80%
Route A (Paris)
Route B (strain)
Cumulative crack growth [nm]
EFC since RPT1 [-]
d)
Route A: Paris law (tension only)
$dl/dt = k_{cr}(T)[\sigma_t(1 - l/R)\, b_{cr}\sqrt{\pi l}\,]^{m_{cr}}$
+
Route B: volume-strain fatigue (contraction only)
$dl/dt = k_V(T)\max(-\dot{\varepsilon}_V, 0)\, l(1 - l/R)$
Miner-linear: $\Delta \ln l \approx k_V \Delta\varepsilon_V$ per contraction
Crack length l ⇒ roughness $\rho = 1 + 2l\, w_{cr}\rho_{cr}$
SEI on the new crack faces: LLI and pore filling
Pores close ⇒ isolation (mechanism I)
Both routes slower when hot: E_{cr} = +90, E_V = +150 kJ mol^{-1}